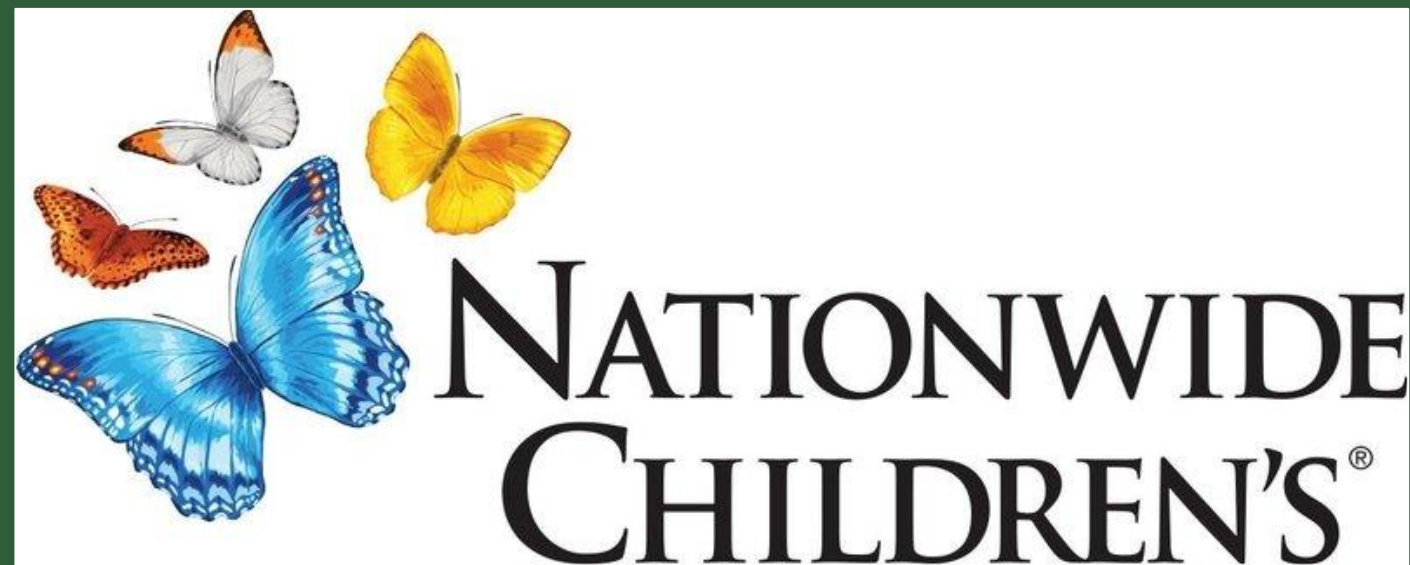


Characteristics of Appalachian Suicide Deaths in Young People: An NVDRS Analysis

Gabrielle B. McColly¹, Margaret V. Meyers², Reagan M. Hicks¹, Jonathan L. Doyle³, Laura J. Chavez, PhD^{1,4}, Yinuo Liu¹, Eric A. Youngstrom, PhD^{1,4}, Danielle L. Steelesmith, PhD^{1,4}

¹Nationwide Children's Hospital ; ²University of North Carolina at Chapel Hill ; ³UC Berkeley School of Public Health; ⁴The Ohio State University College of Medicine

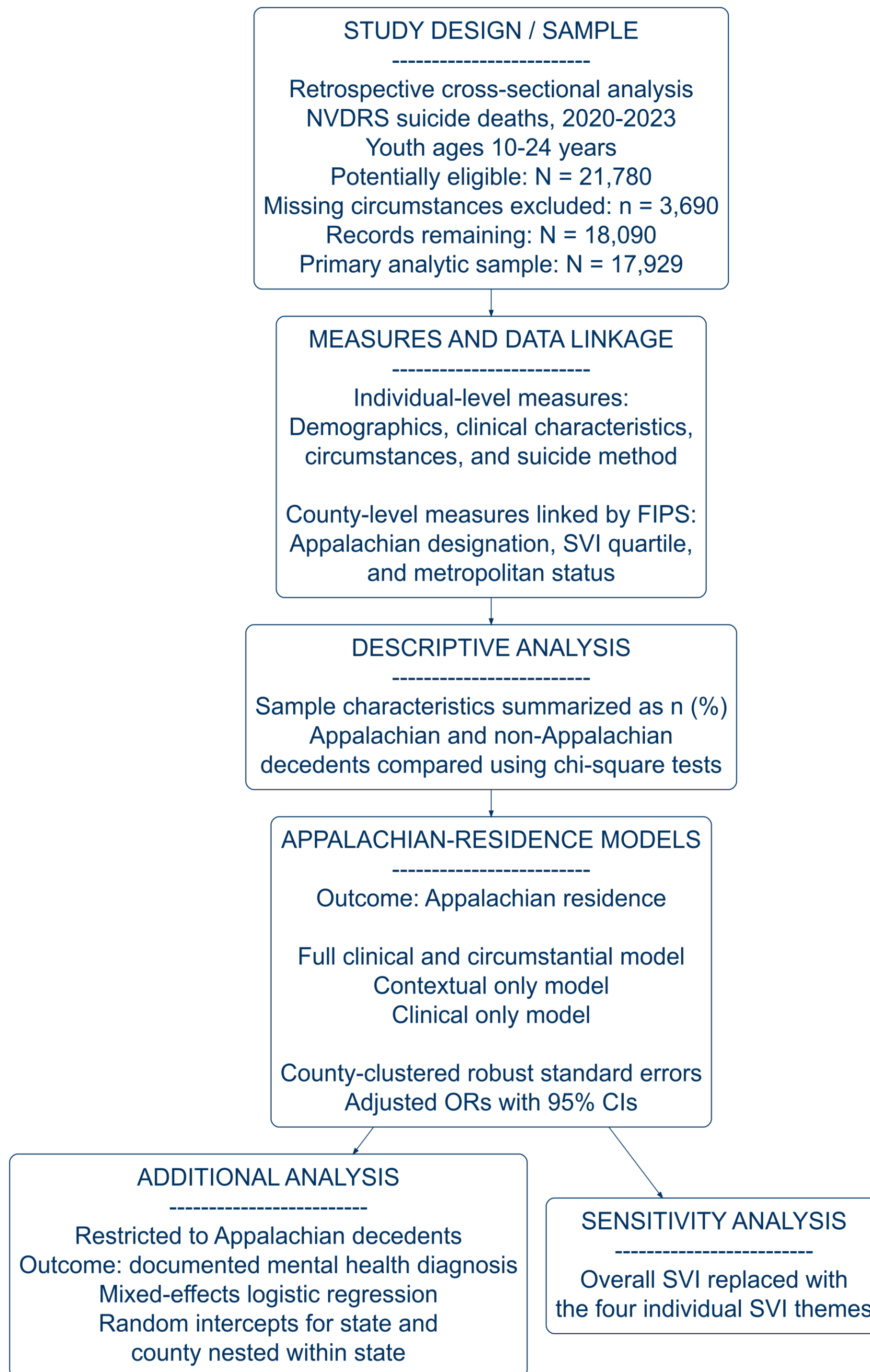


The authors have no disclosures to report.

BACKGROUND

- Suicide is a leading cause of death among young people in the United States¹.
- The Appalachian region experiences unique challenges that may contribute to suicide risk^{2, 3}.
- Despite these disparities, no study has examined youth suicide deaths across all Appalachian counties.
- The current study aimed to address this gap by:
 - 1) developing a comprehensive understanding of the individual and clinical characteristics associated with suicide deaths among young people in Appalachia.
 - 2) characterizing socioeconomic disparities and vulnerability among young suicide decedents in Appalachia.

METHODS



Young suicide decedents in Appalachia were more likely to live in rural, socioeconomically vulnerable communities and less likely to have documented mental health concerns.

Geographic Representation of the Appalachian Region in the United States

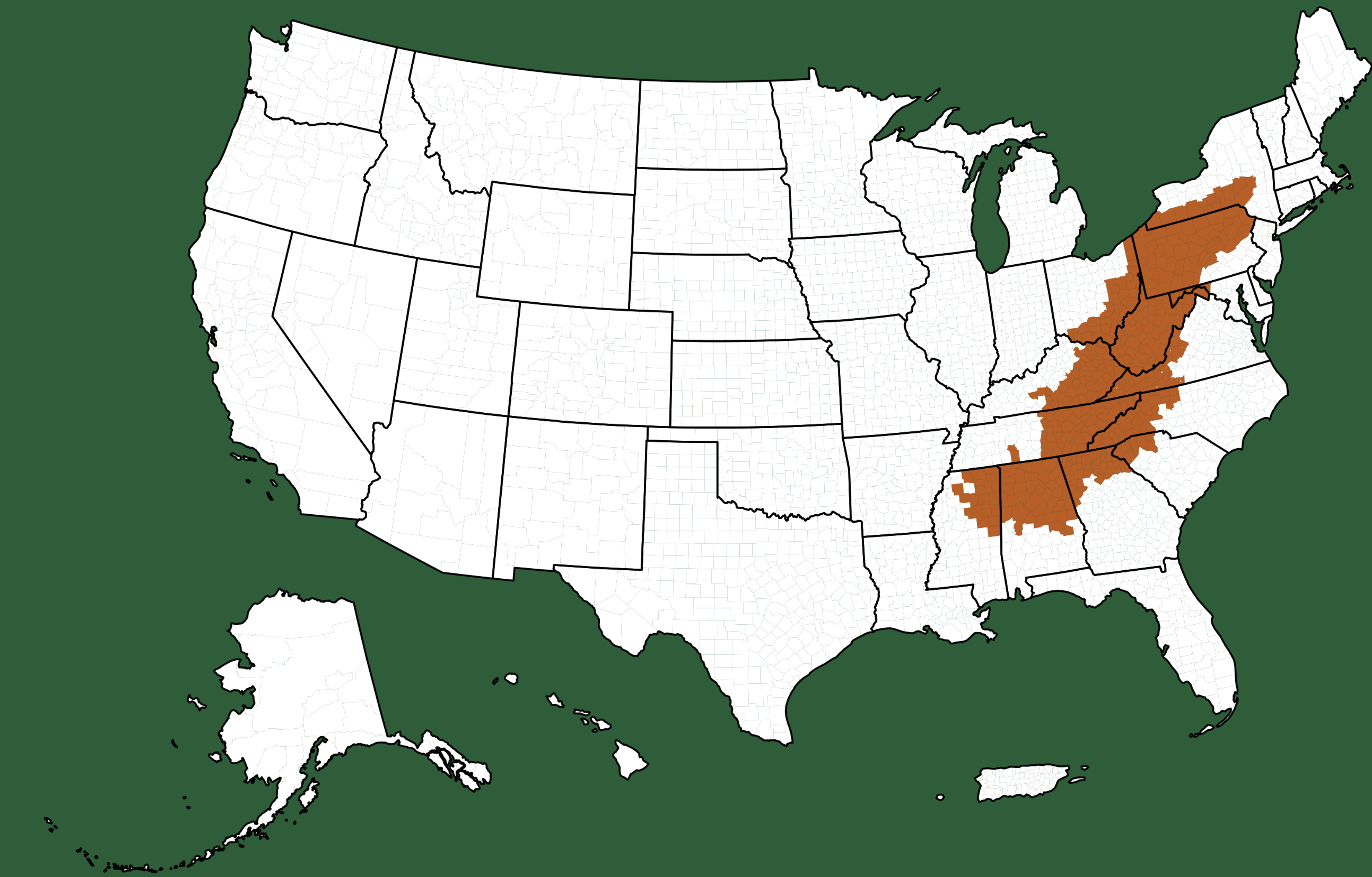


Figure 1. Map of the Appalachian Region. Counties designated as Appalachian are highlighted in orange. The 423 counties span 13 states, and include the entirety of West Virginia.

Scan the QR code for additional project information.



RESULTS

Table 1. Demographic Characteristics by Appalachian Residence

Characteristic	Non-Appalachia (N=16,384)	Appalachia (N=1,545)	p-value
Age, n (%)			.892
10-17 years	4,014 (24.5)	379 (24.5)	
18-24 years	12,370 (75.5)	1,166 (75.5)	
Sex, n (%)			.003
Female	3,715 (22.7)	299 (19.4)	
Male	12,669 (77.3)	1,246 (80.6)	
Race/Ethnicity, n (%)			<.001
Non-Hispanic White	9,378 (57.2)	1,208 (78.2)	
Non-Hispanic Black	2,244 (13.7)	180 (11.7)	
Hispanic	2,988 (18.2)	93 (6.0)	
Other*	1,774 (10.9)	64 (4.1)	

*Includes Non-Hispanic Asian, Non-Hispanic Two or More Races, and Other/Unknown.

Factors Associated with Appalachian Residence Among Young Suicide Decedents

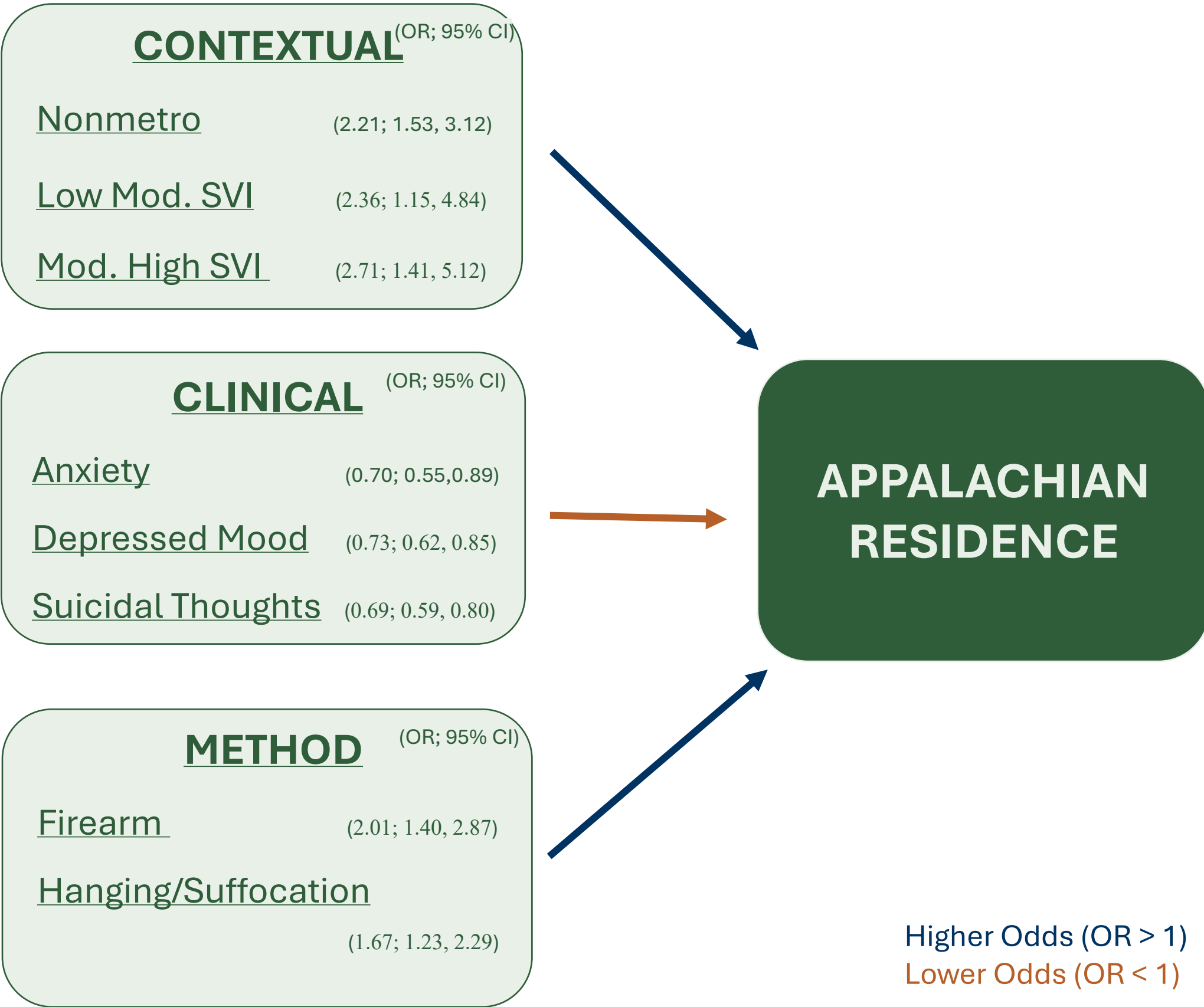


Figure 2. Path diagram illustrating factors associated with Appalachian residence among young suicide decedents. Contextual characteristics and suicide methods were associated with increased odds of residing in Appalachia, while clinical characteristics were associated with decreased odds. Odds ratios and 95% confidence intervals are included.

DISCUSSION

- Appalachian and non-Appalachian young suicide decedents differed across demographic, clinical, and contextual factors.
- Appalachian decedents were more likely to reside in nonmetro, socioeconomically vulnerable communities and less likely to have documented mental health concerns.
- Findings highlight the need to better understand regional influences on suicide to inform community-based prevention efforts.